

Capstone Project Document

AI-powered Film Photography Contest Management Platform	
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We are especially grateful to our supervisor. Your dedicated guidance, valuable insights, and constructive criticism throughout the semester have been instrumental in shaping this project. Your expertise helped us navigate complex challenges, from defining use cases and screen flows to finalizing this SRS document according to industry standards.

Finally, although we have worked hard to ensure accuracy, errors may still exist. We welcome any feedback and suggestions from our lecturers to further improve this project.

Definition and Acronyms

Acronym	Definition	Description
API	Application Programming Interface	A set of rules and protocols that allows different software applications to communicate with each other.
BA	Business Analysis	The practice of enabling change in an enterprise by defining needs and recommending solutions that deliver value to stakeholders.
BR	Business Rule	A rule that defines or constrains some aspect of business and always resolves to either true or false.
ERD	Entity Relationship Diagram	A flowchart that illustrates how "entities" such as people, objects, or concepts relate to each other within a system.

Acronym	Definition	Description
EXIF	Exchangeable Image File Format	A standard format for specifying meta-data tags embedded in digital image files (e.g., Camera Model, Lens, Film Stock/ISO).
GUI	Graphical User Interface	A type of user interface that allows users to interact with electronic devices through graphical icons and visual indicators.
MHs	Man-Hours	An estimated unit of work equal to one hour of continuous labor by one project team member.
pHash	Perceptual Hashing	A fingerprinting algorithm used to analyze image contents and generate a unique hash for duplicate image detection.
PM	Project Manager	The person responsible for planning, executing, monitoring, controlling, and closing projects.
RBAC	Role-Based Access Control	A security approach that restricts system access based on defined user roles (e.g., Candidate, Judge, Organizer).
SDD	Software Design Description	A document that describes the design of a software system, including its architecture, components, and interfaces.
SPMP	Software Project Management Plan	A document that defines the project plan, specifying the tasks, schedule, resources, and management approach.
SRS	Software Requirement Specification	A document that describes what the software will do and how it will be expected to perform.
UAT	User Acceptance Test	The final phase of the software testing process where actual software users test the software to make sure it can handle required tasks in real-world scenarios.
UC	Use Case	A list of actions or event steps typically defining the interactions between a role and a system to achieve a goal.

1 Project Introduction

1.1 Overview

1.1.1 Project Information

Project Name: AI-powered Film Photography Contest Management Platform

Software Type: Web Application

Purpose: To streamline the organization, submission, judging, and exhibition processes of film photography competitions.

Group Name: Group 7

1.2 Project Team

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1.3 Product Background

Film photography has become increasingly popular in recent years, leading to a growing number of photography contests. However, most contests are still managed using separate tools such as online forms, spreadsheets, emails, and cloud storage, resulting in inefficient workflows and difficulties in submission verification, judging, and data management.

Unlike digital photography contests, film photography requires additional technical metadata such as film stock, camera, lens, ISO, developing laboratory, and scanning information. Existing contest management systems generally do not support these specialized requirements.

To address these limitations, this project proposes an AI-powered Film Photography Contest Management Platform. The platform centralizes contest management, supports film-specific metadata, and integrates AI for duplicate image detection, AI-generated image detection, automatic image categorization, and contest analytics, improving efficiency, transparency, and fairness throughout the competition process.

1.4 Existing Systems

1.4.1 Picter

Link: <https://www.picter.com>

Description: A professional platform for managing photography competitions, open calls, grants, and awards. It is widely used by organizations such as World Press Photo, Leica, and Magnum.

Actors: Organizers, judges, photographers, administrators.

Key Features:

- Online contest creation and management.
- Photo submission and portfolio management.
- Online judging and scoring.
- Result publication and digital archive.

Pros: Complete end-to-end contest management; professional judging workflow; supports large-scale international competitions.

Cons: Not designed specifically for film photography; does not provide film-specific metadata management (film stock, camera, developing lab, etc.); limited AI verification features.

1.4.2 CameraComp

Link: <https://www.cameracomp.com>

Description: A cloud-based competition management system developed for photography clubs and federations.

Actors: Club administrators, judges, photographers.

Key Features:

- Competition management.
- Image submission.
- Online and offline judging.
- Member and event management.

Pros: Simple and intuitive interface; integrated competition and member management; supports multiple judging methods.

Cons: Focuses on photography clubs rather than film photography contests; no AI-generated image detection; no film metadata management.

Compared with the systems above, the proposed platform is specifically designed for film photography contests. It focuses on film-specific metadata, AI-based verification, and a more specialized workflow for organizers, judges, and participants.

1.5 Business Opportunity

The growing popularity of film photography has increased the demand for efficient contest management. However, most organizers still use separate tools, making submission management, judging, and verification inefficient. This project provides a centralized platform with AI support to improve efficiency, transparency, and fairness for photography contests.

1.6 Software Product Vision

The project aims to develop a web and mobile platform for managing film photography contests. It supports the complete contest workflow, from registration and submission to judging and result publication, while integrating AI for duplicate image detection, AI-generated image detection, and automated image categorization to enhance contest management.

1.7 Project Scope & Limitations

1.7.1 Major Features

The following primary functional modules are included:

- **FE1 – User & Role Management:** Provides authentication, role-based access control (RBAC), and profile management for Candidates, Judges, and Organizers/Admins.
- **FE2 – Contest Management:** Enables Organizers to create and configure new contests, set submission rules, manage timelines, oversee participant lists, and assign judges.
- **FE3 – Submission System:** Allows candidates to upload high-resolution film scan images, input detailed technical specifications (EXIF metadata, Camera, Lens, Film Stock/ISO, Aperture, Shutter Speed), and track submission statuses.
- **FE4 – Judging & Evaluation:** Provides a dedicated judging panel featuring high-resolution image viewing tools (Zoom/Pan), multi-criteria evaluation scoring, and feedback/commenting capabilities.

- **FE5 – Automated Processing & Anti-cheat:** Automatically extracts EXIF data from uploaded images and utilizes perceptual hashing (pHash) with Hamming distance calculations to detect and flag potential image plagiarism or duplicate submissions.
- **FE6 – Exhibition Gallery:** A public online gallery showcasing winning or selected submissions, equipped with advanced filtering options by Film Stock, Camera model, or Contest.

1.7.2 Limitations & Exclusions

The following points detail the operational constraints and out-of-scope boundaries:

- **LI-01:** File size and format restrictions: Supports only .jpg, .jpeg, and .png formats, with file sizes strictly limited between 2 MB and 25 MB.
- **LI-02:** Automated metadata extraction depends entirely on intact EXIF data within uploaded film scan files; stripped EXIF files require manual input and administrative flagging.
- **LI-03:** Perceptual hashing (pHash) and Hamming distance duplication detection are processed asynchronously via message queues, resulting in slight processing delays rather than real-time feedback upon upload.
- **LI-04:** No support for native mobile applications (iOS/Android); the system is strictly developed as a responsive web application.
- **LI-05:** No integration with online payment gateways or photo editing/post-processing tools (candidates must upload finalized image files, and all financial transactions are out of scope).